DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-400-05 — Replace spool SP04 of 4"-P300-P-1019

| | | | |
|---|---|---|---|
| **Area** | A400 Pumps | **Task** | SR Spool replacement |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | **P-104A** — P-104A centrifugal pump — E 116'-6" N 18'-6" EL 100'-9" | | |
| **Equipment** | P-104A | **Lines** | 4"-P300-P-1019, 3/4"-P300-I-7019, 3/4"-P300-SP-7028 |
| **Isometrics** | ISO-4"-P300-P-1019, ISO-3/4"-P300-I-7019, ISO-3/4"-P300-SP-7028 (generated); owner's drawings ISO-P-1019 | | |
| **Permits** | General work permit, Hot work permit | | |

## Reason

Wall loss below the owner's minimum found by the last survey on the discharge run to V-102.

## Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P300-P-1019 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P300-P-1019-SP04 (12 items, 17'-8-13/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P300-P-1019 with the same items (parts list), welding to WS-01; 11 shop welds.
4. Install the spool; make 4 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 420 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P300-P-1019/el21, 4"-P300-P-1019/p22, 4"-P300-P-1019/el23, 4"-P300-P-1019/p24, 4"-P300-P-1019/T:3/4"-P300-I-7019, 4"-P300-P-1019/p26, 4"-P300-P-1019/T:3/4"-P300-SP-7028, 4"-P300-P-1019/p28, 4"-P300-P-1019/el29, 4"-P300-P-1019/p30, 4"-P300-P-1019/el31, 4"-P300-P-1019/p32

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-P-1019 | 4"-P300-P-1019 | 4"-P300-P-1019 | 5 | 3 error(s), 2 warning(s) — see below |

ISO-P-1019 (PDF) — the generated isometrics follow the ITP.

| Drawing | Sheet | Level | Finding |
|---|---|---|---|
| ISO-P-1019 | 4 | error | sheet 4: CONT'D ON ISO-P-1019 SH 5 prints E 53'-6" N -11'-6" EL 120'-0" for E 53'-6" N -10'-6" EL 120'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off the line's centreline |
| ISO-P-1019 | 5 | error | sheet 5: CONT'D FROM ISO-P-1019 SH 4 prints E 53'-6" N -11'-6" EL 120'-0" for E 53'-6" N -10'-6" EL 120'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off the line's centreline |
| ISO-P-1019 | 5 | error | W22 (4"-P300-P-1019/W22) is labelled on sheet 4 but is on sheet 5, 5.3 in from the break |
| ISO-P-1019 | 3 | warning | branch weld 3/4"-P300-I-7019/W1 (E 53'-6" N 3'-6" EL 119'-3 5/16", sheet 3) has no W27 label |
| ISO-P-1019 | 4 | warning | branch weld 3/4"-P300-SP-7028/W1 (E 53'-6" N 2'-0" EL 119'-3 5/16", sheet 4) has no W28 label |

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| P-104A | equipment | P-104A centrifugal pump | | E 116'-6" N 18'-6" EL 100'-9" |
| 4"-P300-P-1019/el21 | item | ELBOW 90° R = 5/3 OD 4" P300 | 4"-P300-P-1019 | E 53'-9-5/16" N 5'-0" EL 116'-9-5/16" |
| 4"-P300-P-1019/p22 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N 5'-0" EL 117'-9" |
| 4"-P300-P-1019/el23 | item | ELBOW 90° R = 5/3 OD 4" P300 | 4"-P300-P-1019 | E 53'-6" N 4'-8-11/16" EL 118'-8-11/16" |
| 4"-P300-P-1019/p24 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N 4'-1-5/16" EL 119'-0" |
| 4"-P300-P-1019/T:3/4"-P300-I-7019 | item | TEE REDUCING 4" × 3/4" P300 | 4"-P300-P-1019 | E 53'-6" N 3'-6" EL 119'-0" |
| 4"-P300-P-1019/p26 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N 2'-9" EL 119'-0" |
| 4"-P300-P-1019/T:3/4"-P300-SP-7028 | item | TEE REDUCING 4" × 3/4" P300 | 4"-P300-P-1019 | E 53'-6" N 2'-0" EL 119'-0" |
| 4"-P300-P-1019/p28 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N -4'-1-5/16" EL 119'-0" |
| 4"-P300-P-1019/el29 | item | ELBOW 90° R = 5/3 OD 4" P300 | 4"-P300-P-1019 | E 53'-6" N -10'-2-11/16" EL 119'-3-5/16" |

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| 4"-P300-P-1019/p30 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N -10'-6" EL 120'-0" |
| 4"-P300-P-1019/el31 | item | ELBOW 90° R = 5/3 OD 4" P300 | 4"-P300-P-1019 | E 53'-6" N -10'-9-5/16" EL 120'-8-11/16" |
| 4"-P300-P-1019/p32 | item | PIPE 4" P300 BEVELLED ENDS | 4"-P300-P-1019 | E 53'-6" N -12'-1-3/4" EL 121'-0" |
| 4"-P300-P-1019/A5 | item | VALVE, GATE, WEDGE, OS&Y 3" P300 | 4"-P300-P-1019 | E 116'-6" N 16'-6" EL 105'-2-3/16" |
| 4"-P300-P-1019/A8 | item | FLANGE WELD-NECK 3" P300 RAISED FACE | 4"-P300-P-1019 | E 116'-6" N 16'-6" EL 105'-11-5/8" |
| 4"-P300-P-1019/B16 | item | VALVE, GATE, WEDGE, OS&Y 4" P300 | 4"-P300-P-1019 | E 53'-6" N -14'-4" EL 121'-0" |
| 4"-P300-P-1019/B19 | item | FLANGE WELD-NECK 4" P300 RAISED FACE | 4"-P300-P-1019 | E 53'-6" N -13'-5-1/2" EL 121'-0" |
| 4"-P300-P-1019/W12 | weld | Butt weld 4" P300 (field) | 4"-P300-P-1019 | E 54'-11/16" N 5'-0" EL 116'-6" |
| 4"-P300-P-1019/W13 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 5'-0" EL 117'-11/16" |
| 4"-P300-P-1019/W14 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 5'-0" EL 118'-5-5/16" |
| 4"-P300-P-1019/W15 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 4'-5-5/16" EL 119'-0" |
| 4"-P300-P-1019/W16 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 3'-9-5/16" EL 119'-0" |
| 4"-P300-P-1019/W17 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 3'-2-11/16" EL 119'-0" |
| 4"-P300-P-1019/W18 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 2'-3-5/16" EL 119'-0" |
| 4"-P300-P-1019/W19 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N 20-11/16" EL 119'-0" |
| 4"-P300-P-1019/W20 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N -9'-11-5/16" EL 119'-0" |
| 4"-P300-P-1019/W21 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N -10'-6" EL 119'-6-11/16" |
| 4"-P300-P-1019/W22 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N -10'-6" EL 120'-5-5/16" |
| 4"-P300-P-1019/W23 | weld | Butt weld 4" P300 (shop) | 4"-P300-P-1019 | E 53'-6" N -11'-11/16" EL 121'-0" |
| 4"-P300-P-1019/W24 | weld | Butt weld 4" P300 (field) | 4"-P300-P-1019 | E 53'-6" N -13'-2-13/16" EL 121'-0" |
| 3/4"-P300-I-7019/W1 | weld | Butt weld 0.75" P300 (field) | 3/4"-P300-I-7019 | E 53'-6" N 3'-6" EL 119'-3-5/16" |
| 3/4"-P300-SP-7028/W1 | weld | Butt weld 0.75" P300 (field) | 3/4"-P300-SP-7028 | E 53'-6" N 2'-0" EL 119'-3-5/16" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 4"-P300-P-1019/A5 | 4"-P300-P-1019 | Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1019/A5) | E 116'-6" N 16'-6" EL 105'-2-3/16" | | |
| 2 | blind | 4"-P300-P-1019/A8 | 4"-P300-P-1019 | Fit a line blind between 4"-P300-P-1019/A5 and FLANGE WELD-NECK 3" P300 RAISED FACE (4"-P300-P-1019/A8), work side | E 116'-6" N 16'-6" EL 105'-9-1/16" | | |
| 3 | valve | 4"-P300-P-1019/B16 | 4"-P300-P-1019 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1019/B16) | E 53'-6" N -14'-4" EL 121'-0" | | |
| 4 | blind | 4"-P300-P-1019/B19 | 4"-P300-P-1019 | Fit a line blind between 4"-P300-P-1019/B16 and FLANGE WELD-NECK 4" P300 RAISED FACE (4"-P300-P-1019/B19), work side | E 53'-6" N -13'-8-1/4" EL 121'-0" | | |

## Parts

| Description | Qty | Replaces |
|---|---|---|
| ELBOW 90° R = 5/3 OD 4" P300 | 4 | |
| PIPE 4" P300 BEVELLED ENDS | 6 | |
| TEE REDUCING 4" × 3/4" P300 | 2 | |

## New welds (SW-WO-4)

| Weld | Replaces | On drawing | Line | Size | Wall | Shop/field | Spec | NDE | Location | Welder | Date |
|---|---|---|---|---|---|---|---|---|---|---|---|
| WO-400-05-W01 | 4"-P300-P-1019/W12 | ISO-P-1019 sh 2 W12 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | field | WS-01 | VT + RT | E 54'-11/16" N 5'-0" EL 116'-6" | | |
| WO-400-05-W02 | 4"-P300-P-1019/W13 | ISO-P-1019 sh 2 W13 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 5'-0" EL 117'-11/16" | | |
| WO-400-05-W03 | 4"-P300-P-1019/W14 | ISO-P-1019 sh 3 W14 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 5'-0" EL 118'-5-5/16" | | |
| WO-400-05-W04 | 4"-P300-P-1019/W15 | ISO-P-1019 sh 3 W15 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 4'-5-5/16" EL 119'-0" | | |
| WO-400-05-W05 | 4"-P300-P-1019/W16 | ISO-P-1019 sh 3 W16 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 3'-9-5/16" EL 119'-0" | | |
| WO-400-05-W06 | 4"-P300-P-1019/W17 | ISO-P-1019 sh 3 W17 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 3'-2-11/16" EL 119'-0" | | |

| Weld | Replaces | On drawing | Line | Size | Wall | Shop/field | Spec | NDE | Location | Welder | Date |
|---|---|---|---|---|---|---|---|---|---|---|---|
| WO-400-05-W07 | 4"-P300-P-1019/W18 | ISO-P-1019 sh 4 W18 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 2'-3-5/16" EL 119'-0" | | |
| WO-400-05-W08 | 4"-P300-P-1019/W19 | ISO-P-1019 sh 4 W19 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N 20-11/16" EL 119'-0" | | |
| WO-400-05-W09 | 4"-P300-P-1019/W20 | ISO-P-1019 sh 4 W20 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N -9'-11-5/16" EL 119'-0" | | |
| WO-400-05-W10 | 4"-P300-P-1019/W21 | ISO-P-1019 sh 4 W21 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N -10'-6" EL 119'-6-11/16" | | |
| WO-400-05-W11 | 4"-P300-P-1019/W22 | ISO-P-1019 sh 5 W22 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N -10'-6" EL 120'-5-5/16" | | |
| WO-400-05-W12 | 4"-P300-P-1019/W23 | ISO-P-1019 sh 5 W23 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | shop | WS-01 | VT + RT | E 53'-6" N -11'-11/16" EL 121'-0" | | |
| WO-400-05-W13 | 4"-P300-P-1019/W24 | ISO-P-1019 sh 5 W24 | 4"-P300-P-1019 | 4" P300 | 0.1875 in | field | WS-01 | VT + RT | E 53'-6" N -13'-2-13/16" EL 121'-0" | | |
| WO-400-05-W14 | 3/4"-P300-I-7019/W1 | ISO-P-1019 sh 3 (no label; W27 by numbering) | 3/4"-P300-I-7019 | 0.75" P300 | 0.09375 in | field | WS-01 | VT + RT | E 53'-6" N 3'-6" EL 119'-3-5/16" | | |
| WO-400-05-W15 | 3/4"-P300-SP-7028/W1 | ISO-P-1019 sh 4 (no label; W28 by numbering) | 3/4"-P300-SP-7028 | 0.75" P300 | 0.09375 in | field | WS-01 | VT + RT | E 53'-6" N 2'-0" EL 119'-3-5/16" | | |

## Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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| | |
|---|---|
| **Scope** | Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in |
| **Joint** | Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4) |
| **Process** | Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc. |
| **Filler** | Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen) |
| **Preheat** | ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side |
| **Interpass** | ≤ 450 °F |
| **Post-weld heat treatment** | Not specified (owner to state) |
| **Rules** | SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4 |

### NDE

| Method | Extent | Welds | Basis |
|---|---|---|---|
| VT | 100 % of new welds, before and after welding | WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15 | SW-NDE-1 |
| RT (UT where RT is impractical) | 100 % of new welds (class H300, high consequence) | WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15 | SW-NDE-2 |

## Test — Hydrostatic test

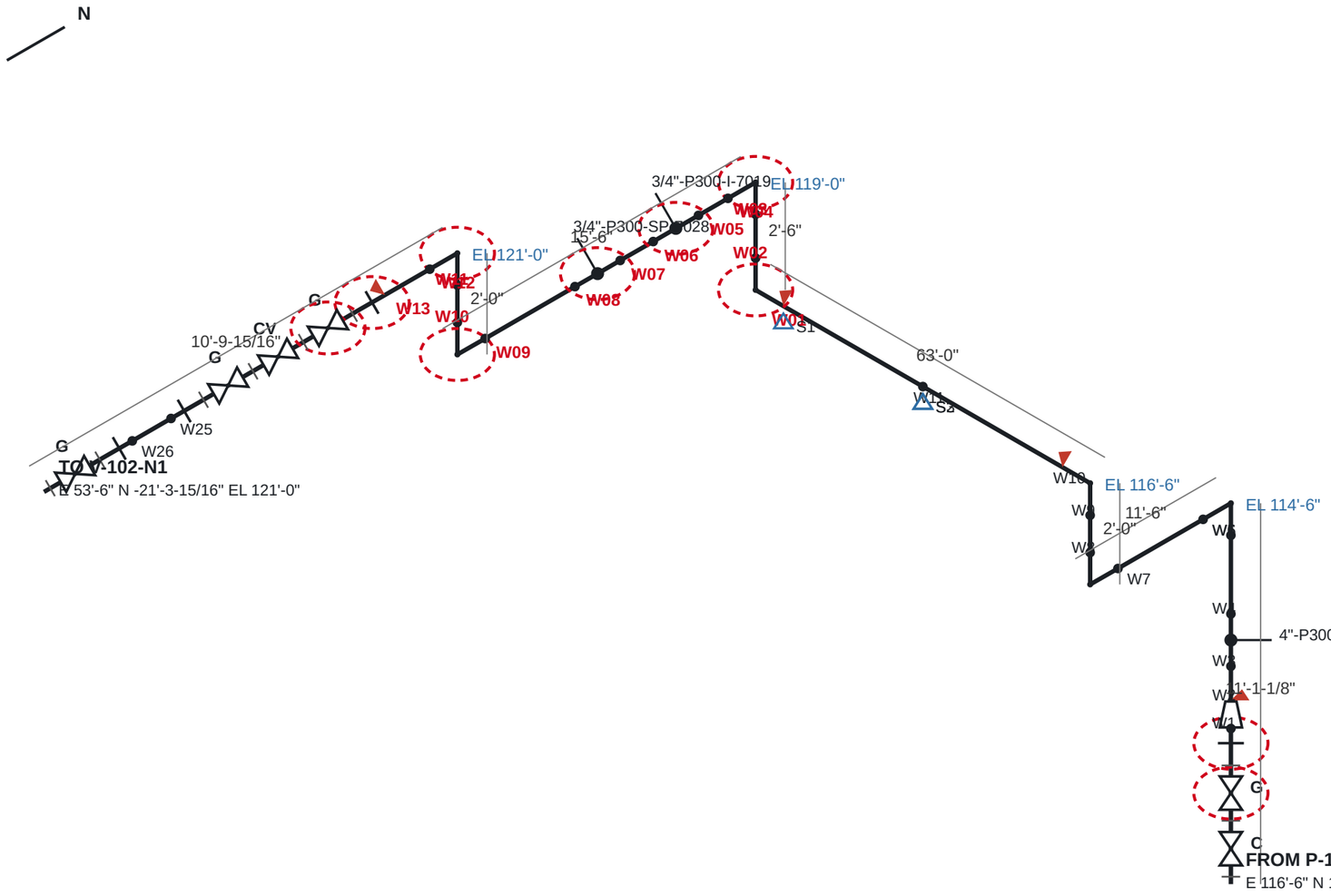
| | |
|---|---|
| **Description** | Test boundary between the isolations; relief valves blinded or removed |
| **Pressure** | 420 psi |
| **Hold** | 30 minutes, no leak, no pressure drop |
| **Basis** | SW-TEST-1 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Line drained, vented and purged; gas-free for hot work | Owner hot-work procedure | Gas test within owner limits | Gas test record, hot work permit | H | H | — | | | | |
| 3 | Replacement materials verified against the parts list (size, class, grade, heat marking) | Parts list, SW-LC-3 | All items match the line class | Material receiving record | R | R | R | | | | |
| 4 | Welding specification WS-01 and welder identification available at the work | WS-01, SW-WPS-3 | Specification on site; every welder identified | Welder list | R | R | R | | | | |
| 5 | Fit-up of every new weld: root opening and alignment within the root face | SW-WPS-3, SW-WD-4 | Root opening 1/8 in; bores aligned within 1/16 in | Fit-up record per weld | H | H | S | | | | |
| 6 | Preheat and interpass temperatures maintained during welding | WS-01, SW-WPS-2 | Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F | Weld record per weld (welder, temperatures) | S | S | S | | | | |
| 7 | Visual examination of every new weld | SW-NDE-1, SW-NDE-4 | No crack, lack of fusion or incomplete penetration | VT report | W | W | W | | | | |
| 8 | Volumetric examination per the NDE table; repairs re-examined | SW-NDE-2, SW-NDE-4 | Acceptance per SW-NDE-4 and the owner's criteria | RT/UT reports | R | H | R | | | | |
| 9 | Flanged ends: faces inspected, new gaskets, studs to preload | SW-NDE-5, SW-WO-5 | No radial scratch; preload as stated | Bolt-up record | W | W | — | | | | |
| 10 | Hydrostatic test at 420 psi, 30 minutes | SW-TEST-1 | No leak, no pressure drop | Test record with gauge calibration | H | H | H | | | | |
| 11 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 12 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

## ISOMETRIC — 4"-P300-P-1019 — Replace spool SP04 of 4"-P300-P-1019




### BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3" P300 COMPRESSED SHEET | 3 |
| 2 | VALVE, SWING CHECK 3" P300 | 1 |
| 3 | STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300 | 3 |
| 4 | VALVE, GATE, WEDGE, OS&Y 3" P300 | 1 |
| 5 | FLANGE WELD-NECK 3" P300 RAISED FACE | 1 |
| 6 | REDUCER CONCENTRIC 4" × 3" P300 | 1 |
| 7 | PIPE 4" P300 BEVELLED ENDS | 99'-2-15/16" |
| 8 | TEE EQUAL 4" P300 | 1 |
| 9 | ELBOW 90° R = 5/3 OD 4" P300 | 7 |
| 10 | TEE REDUCING 4" × 3/4" P300 | 2 |
| 11 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 | 6 |
| 12 | FLANGE WELD-NECK 4" P300 RAISED FACE | 3 |
| 13 | GASKET 4" P300 COMPRESSED SHEET | 6 |
| 14 | VALVE, GATE, WEDGE, OS&Y 4" P300 | 3 |
| 15 | VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 | 1 |

### WELDS 26 (4 FIELD) · SUPPORTS 3 · SPOOLS 6

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22 W23 W24F W25 W26

### NEW WELDS (WO-400-05)

WO-400-05-W01 replaces W12
WO-400-05-W02 replaces W13
WO-400-05-W03 replaces W14
WO-400-05-W04 replaces W15
WO-400-05-W05 replaces W16
WO-400-05-W06 replaces W17
WO-400-05-W07 replaces W18
WO-400-05-W08 replaces W19
WO-400-05-W09 replaces W20
WO-400-05-W10 replaces W21
WO-400-05-W11 replaces W22
WO-400-05-W12 replaces W23
WO-400-05-W13 replaces W24

## 4"-P300-P-1019

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-104A-D → V-102-N1
PIPE 99'-2-15/16" · WEIGHT 1884 lb

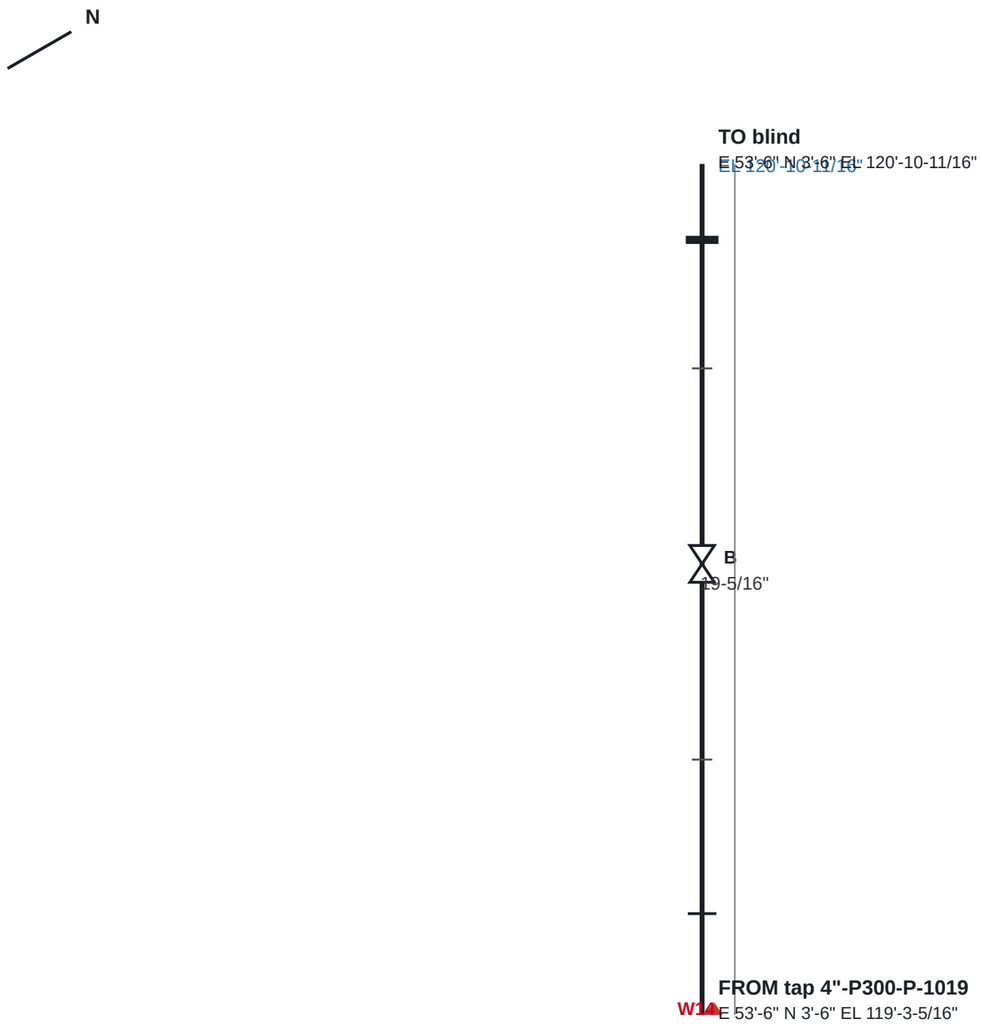
**DWG ISO-4"-P300-P-1019 REV 0 WO-400-05**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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## Isometric ISO-3/4"-P300-I-7019

ISOMETRIC — 3/4"-P300-I-7019 — Replace spool SP04 of 4"-P300-P-1019




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 3/4" P300 RAISED FACE | 1 |
| 2 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 3 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 4 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 5 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 1 (1 FIELD) · SUPPORTS 0 · SPOOLS 1**

W1F

**NEW WELDS (WO-400-05)**

WO-400-05-W14 replaces W1

## 3/4"-P300-I-7019

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
tap 4"-P300-P-1019 → blind
PIPE 0.000" · WEIGHT 37 lb

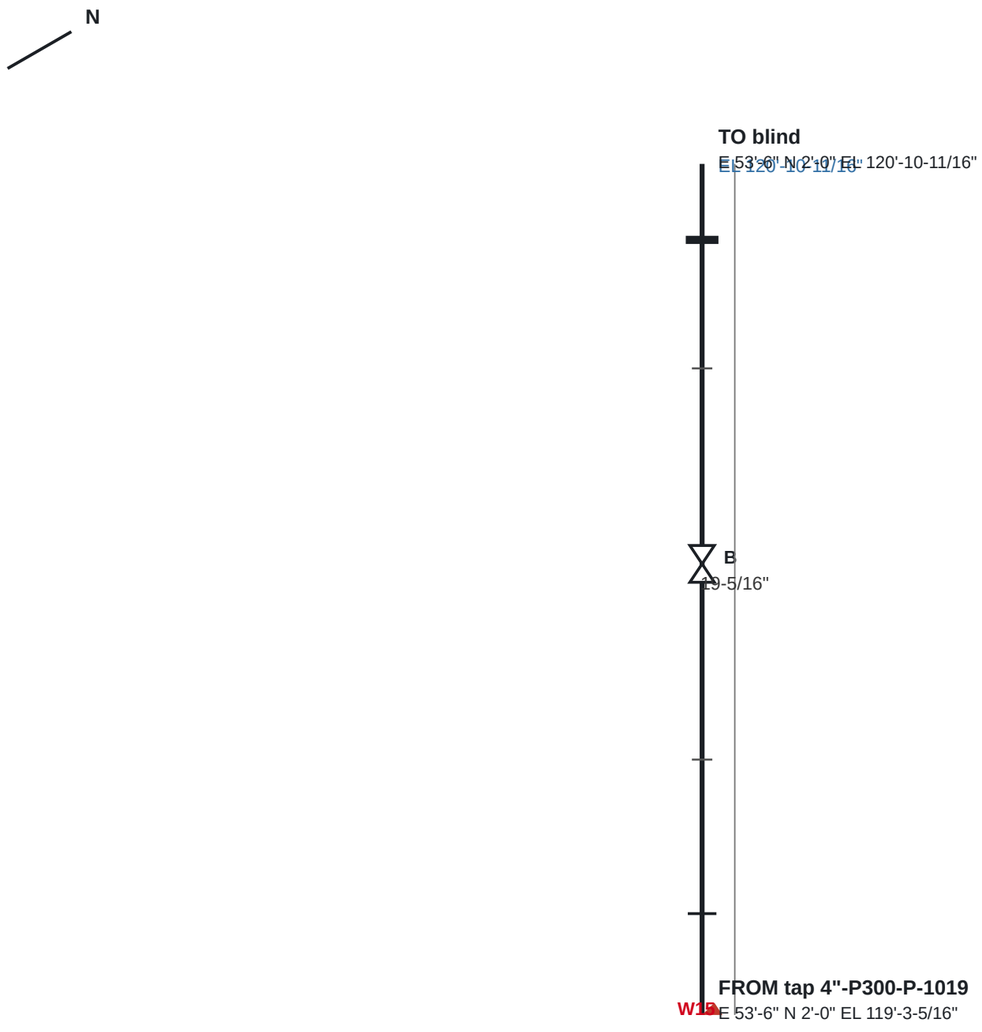
**DWG ISO-3/4"-P300-I-7019 REV 0 WO-400-05**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

## Isometric ISO-3/4"-P300-SP-7028

ISOMETRIC — 3/4"-P300-SP-7028 — Replace spool SP04 of 4"-P300-P-1019




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 3/4" P300 RAISED FACE | 1 |
| 2 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 3 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 4 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 5 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 1 (1 FIELD) · SUPPORTS 0 · SPOOLS 1**

W1F

**NEW WELDS (WO-400-05)**

WO-400-05-W15 replaces W1

## 3/4"-P300-SP-7028

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
tap 4"-P300-P-1019 → blind
PIPE 0.000" · WEIGHT 37 lb

**DWG ISO-3/4"-P300-SP-7028 REV 0 WO-400-05**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**